

# Alzheimer's Disease and Related Dementias (ADRD) Identification

## Julie Bynum 1-Year Standard Method

### General Notes

This README file accompanies the SAS and Stata scripts for the 1-Year Standard Method for identifying ADRD in Medicare Claims data.

There are seven script files (plus a parameters file for SAS [parm.sas]) for both SAS and Stata. The files are numbered in the order in which they should be run; the five "1" files may be run in any order.

The full algorithm requires access to a single year of Medicare Claims data for (1) MedPAR, (2) Home Health Agency (HHA) Claims File, (3) Hospice Claims File, (4) Carrier Claims and Line Files, and (5) Hospital Outpatient File (HOF) Claims and Revenue Files. All Medicare Claims files are expected to be in SAS format (.sas7bdat).

For each data source, the script will output three files\*:

1. Diagnosis-level file: Lists individual ADRD diagnoses for each beneficiary for a given visit. This file allows researchers to identify which ICD-9-CM or ICD-10-CM codes are used in the claims data.
2. Service Date-level file: Aggregated from the Diagnosis-level file, this file includes all beneficiaries with an ADRD diagnosis by Service Date (date of a claim with at least one ADRD diagnosis).
3. Beneficiary-level file: Aggregated from the Service Date-level file, this file includes all beneficiaries with at least one\* ADRD diagnosis at any point in the year within a specific file

\* The algorithm combines the Carrier and HOF files at the Service Date-level. The final combined Carrier and HOF Beneficiary-level file includes those with at least two (2) claims that are seven (7) or more days apart.

A final combined file is created by merging all Beneficiary-level files. This file is used to identify beneficiaries with ADRD and can be merged onto other files by the Beneficiary ID (BENE\_ID).

### Instructions

Each file includes a section at the beginning where users must update a series of path names and macro/local variables to fit your specific dataset and file structure.

Code was written assuming that all Medicare Claims files are .sas7bdat format. If your data is not in this format, you may need to make additional edits to read in your file and ensure that field types match the code.

## Output

### Diagnosis-Level File – “DIAG\_ADRD\_[data\_source]\_[YEAR]” (.sas7bdat or .dta)

| Variable                                 | Type        | Description                     |
|------------------------------------------|-------------|---------------------------------|
| BENE_ID                                  | String/Char | Beneficiary ID                  |
| ServiceDt (SERVICE_DT in Stata)          | Num/Date    | Date of ADRD Dx claim           |
| ServiceThruDt (SERVICE_THRU_DT in Stata) | Num/Date    | Thru date of ADRD Dx claim      |
| ADRD_DX                                  | Num         | Presence of ADRD diagnosis. 1/0 |
| DX                                       | STRING/CHAR | ADRD ICD9/10 Diagnosis code     |

### Service Date-Level File – “SERVDT\_ADRD\_[data\_source]\_[YEAR]” (.sas7bdat or .dta)

| Variable                                 | Type        | Description                     |
|------------------------------------------|-------------|---------------------------------|
| BENE_ID                                  | String/Char | Beneficiary ID                  |
| ServiceDt (SERVICE_DT in Stata)          | Num/Date    | Date of ADRD Dx claim           |
| ServiceThruDt (SERVICE_THRU_DT in Stata) | Num/Date    | Thru date of ADRD Dx claim      |
| ADRD_DX                                  | Num         | Presence of ADRD diagnosis. 1/0 |
| NUM_ADRD_DX                              | Num         | Total ADRD Dx. by Service Date  |

### Beneficiary-Level File – “BENE\_ADRD\_[data\_source]\_[YEAR]” (.sas7bdat or .dta)

| Variable                | Type        | Description                                             |
|-------------------------|-------------|---------------------------------------------------------|
| BENE_ID                 | String/Char | Beneficiary ID                                          |
| ADRD_DX                 | Num         | Presence of ADRD diagnosis. 1/0                         |
| [data_source]_ADRD_DX   | Num         | In [data_source], presence of ADRD diagnosis. 1/0       |
| [data_source]_ADRD_CLMS | Num         | Total number of ADRD claims by bene_id in [data_source] |

**Final Beneficiary-Level File with all sources – “BENE\_ADRD\_ALL\_SOURCES\_[YEAR]” (.sas7bdat or .dta)**

| <b>Variable</b>       | <b>Type</b> | <b>Description</b>                                    |
|-----------------------|-------------|-------------------------------------------------------|
| BENE_ID               | String/Char | Beneficiary ID                                        |
| ADRD_DX               | Num         | Presence of ADRD diagnosis. 1/0                       |
| HOF_CARRIER_ADRD_DX   | Num         | In HOF/Carrier, presence of ADRD diagnosis. 1/0       |
| HOF_CARRIER_ADRD_CLMS | Num         | Total number of ADRD claims by bene_id in HOF/Carrier |
| MEDPAR_ADRD_DX        | Num         | In MedPAR, presence of ADRD diagnosis. 1/0            |
| MEDPAR_ADRD_CLMS      | Num         | Total number of ADRD claims by bene_id in MedPAR      |
| HHA_ADRD_DX           | Num         | In HHA, presence of ADRD diagnosis. 1/0               |
| HHA_ADRD_CLMS         | Num         | Total number of ADRD claims by bene_id in HHA         |
| HOSPICE_ADRD_DX       | Num         | In Hospice, presence of ADRD diagnosis. 1/0           |
| HOSPICE_ADRD_CLMS     | Num         | Total number of ADRD claims by bene_id in Hospice     |